

EXPERIMENT – 1

Title: SQL*Plus Formatting Commands

Aim / Objective

To study and implement various **SQL*Plus formatting commands** for displaying query results in a well-organized and readable format.

Theory

SQL*Plus is an Oracle command-line interface that allows users to execute SQL and PL/SQL statements. It also provides several **formatting commands** to improve the appearance of query outputs without modifying the actual data stored in the database.

Formatting commands are useful for generating reports, improving readability, and presenting data in a structured manner.

SQL*Plus formatting commands affect only the display of data and do not alter the database contents.

Common SQL*Plus Formatting Commands

Command	Purpose
COLUMN	Formats column display
HEADING	Changes column heading
FORMAT	Specifies display format
SET LINESIZE	Sets output line width
SET PAGESIZE	Sets number of rows per page
SET VERIFY	Displays substitution variable values
SET FEEDBACK	Displays number of rows returned
SET NULL	Displays text instead of NULL values
TTITLE	Displays title at the top of each page
BTITLE	Displays title at the bottom of each page
BREAK	Suppresses repeated values
COMPUTE	Performs calculations such as SUM, AVG, COUNT

Learning Objectives

After completing this experiment, students will be able to:

- Understand SQL*Plus formatting commands.
- Format query output.
- Change column headings.
- Display titles and page settings.
- Generate professional-looking reports.

Software Requirements

- Oracle Database
- Oracle SQL*Plus / Oracle SQL Developer

Algorithm / Procedure

1. Open SQL*Plus or Oracle SQL Developer.
2. Connect to the Oracle database.
3. Create the Employee table.
4. Insert sample records.
5. Execute formatting commands.
6. Observe the formatted output.
7. Stop.

Program

Step 1 : Create Employee Table

```
CREATE TABLE Employee  
(  
  EmpID NUMBER PRIMARY KEY,  
  EmpName VARCHAR2(30),  
  Department VARCHAR2(30),  
  Salary NUMBER  
);
```

Step 2 : Insert Records

```
INSERT INTO Employee VALUES(101,'Rahul','IT',35000);
```

```
INSERT INTO Employee VALUES(102,'Priya','HR',42000);
```

```
INSERT INTO Employee VALUES(103,'Aman','Finance',39000);
```

```
INSERT INTO Employee VALUES(104,'Sneha','Marketing',45000);
```

```
COMMIT;
```

Program 1 : Display Employee Table

```
SELECT * FROM Employee;
```

Program 2 : Change Column Heading

```
COLUMN EmpName HEADING 'Employee Name'
```

```
SELECT EmpName,Salary  
FROM Employee;
```

Expected Output

Employee Name	Salary
Rahul	35000
Priya	42000
Aman	39000
Sneha	45000

Program 3 : Format Salary Column

```
COLUMN Salary FORMAT 999,999;
```

```
SELECT EmpName,Salary  
FROM Employee;
```

Expected Output

Employee Name Salary

Rahul	35,000
Priya	42,000
Aman	39,000
Sneha	45,000

Program 4 : Set Line Size

```
SET LINESIZE 120;
```

```
SELECT * FROM Employee;
```

Program 5 : Set Page Size

```
SET PAGESIZE 20;
```

```
SELECT * FROM Employee;
```

Program 6 : Display Title

```
TTITLE CENTER 'EMPLOYEE REPORT'
```

```
SELECT * FROM Employee;
```

Expected Output

EMPLOYEE REPORT

EmpID	Employee Name	Department	Salary
101	Rahul	IT	35000
102	Priya	HR	42000
103	Aman	Finance	39000
104	Sneha	Marketing	45000

Program 7 : Display Bottom Title

```
BTITLE CENTER 'End of Report'
```

```
SELECT * FROM Employee;
```

Program 8 : Display NULL Value

```
SET NULL 'Not Available';
```

```
SELECT *  
FROM Employee;
```

Program 9 : BREAK Command

```
BREAK ON Department;
```

```
SELECT Department,  
EmpName,  
Salary  
FROM Employee;
```

Program 10 : COMPUTE Command

BREAK ON Department;

COMPUTE SUM OF Salary ON Department;

SELECT Department,
EmpName,
Salary
FROM Employee;

Sample Output

Department	Employee Name	Salary
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Finance

	Aman	39000

Sum	39000	

HR

	Priya	42000

Sum	42000	

IT

	Rahul	35000

Sum	35000	

Explanation

COLUMN

Formats column headings and display.

FORMAT

Changes the display format of numeric values.

SET LINESIZE

Specifies the width of the output.

SET PAGESIZE

Defines the number of rows displayed on one page.

TTITLE

Displays a report title.

BTITLE

Displays a footer.

BREAK

Suppresses duplicate values.

COMPUTE

Calculates SUM, AVG, MIN, MAX, and COUNT.

SET NULL

Displays custom text in place of NULL values.

Observation

- SQL*Plus formatting commands improved the readability of query results.
 - Column headings were customized successfully.
 - Numeric values were formatted correctly.
 - Report titles and page settings were applied successfully.
 - Aggregate calculations were displayed using the COMPUTE command.
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Result

SQL*Plus formatting commands were successfully executed. Query outputs were formatted using **COLUMN, FORMAT, TTITLE, BTITLE, BREAK, COMPUTE, SET LINESIZE, SET PAGESIZE, and SET NULL**, resulting in professional and readable reports.

Applications

- Financial Reports
 - Payroll Reports
 - Student Result Reports
 - Employee Reports
 - Inventory Reports
 - Banking Reports
 - Sales Reports
-

Advantages

- Improves readability of reports.
 - Generates professional output.
 - Easy customization of query results.
 - Useful for printing reports.
 - Does not modify database contents.
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Viva Questions

1. What is SQL*Plus?
2. What is the purpose of SQL*Plus formatting commands?
3. What is the difference between SQL and SQL*Plus?
4. What is the use of the COLUMN command?
5. What is the purpose of the FORMAT command?
6. What is the use of SET LINESIZE?
7. What is the purpose of TTITLE and BTITLE?
8. What is the BREAK command?

9. What is the COMPUTE command?
 10. Do formatting commands change the data stored in the database?
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Lab Assignment

1. Create a Student table and apply SQL*Plus formatting commands.
2. Change the heading of the Student Name column.
3. Format the Marks column with commas.
4. Display a report title using TTITLE.
5. Display a footer using BTITLE.
6. Use BREAK and COMPUTE to display department-wise salary totals.
7. Display "N/A" instead of NULL values.
8. Change the page size and line size.
9. Generate a formatted Employee Report.
10. Explain the purpose of each SQL*Plus formatting command with examples.

Precautions

- Execute *SQLPlus formatting commands* in ****Oracle SQLPlus**** or **Oracle SQL Developer**.
- Ensure that the table contains data before applying formatting commands.
- Use appropriate numeric formats for number columns.
- Reset formatting commands if required using the CLEAR command.
- Remember that formatting commands affect only the display and not the stored data.

Learning Outcome

After completing this experiment, students will be able to:

- Use SQL*Plus formatting commands to customize query output.
- Format column headings and numeric values.
- Generate professional reports with titles and summaries.
- Improve the readability of SQL query results.
- Apply SQL*Plus reporting features in practical database applications.